Day 057 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Combine Like Terms

Standards: 6.EE.A.3

Learning target: I can combine terms with the same variable part.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Terms with the same variable part count the same kind of object. Add their coefficients. Keep constants separate from variable terms and distribute before combining when parentheses are present.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 16

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01 / Learn

Terms with the same variable part count the same kind of object. Add their coefficients. Keep constants separate from variable terms and distribute before combining when parentheses are present.

Worked example 1

Simplify 8x + 4x. Explain which terms combine.

1. Terms with the same variable part count the same kind of object.

2. Add their coefficients.

3. Add coefficients of x terms

4. add only constants to constants.

5. Distribute first if needed.

6. State the result in the form requested: 12x.

Answer: 12x

Worked example 2

Select all expressions equivalent to 6x + 6x + 3.
A. 12x + 3
B. 12x + 4
C. 15x
D. (6 + 6)x + 3

1. Terms with the same variable part count the same kind of object.

2. Combine coefficients of like x terms: 6 + 6 = 12.

3. The constant 3 stays separate.

Answer: A, D - 12x + 3; (6 + 6)x + 3

Day 057 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Simplify 6x + 3x. Explain which terms combine.

02 / Constructed response

Simplify 4x + 3x. Explain which terms combine.

03 / Constructed response

Simplify 3x + 4x. Explain which terms combine.

04 / Constructed response

Simplify 4x + 3 + 4x + 4. Explain which terms combine.

05 / Constructed response

Simplify 2x + 6 + 7x + 2. Explain which terms combine.

06 / Constructed response

Simplify 4x + 3 + 3x + 4. Explain which terms combine.

07 / Constructed response

Simplify 2(x + 5) + 3x. Explain which terms combine.

08 / Multiple select

Select all expressions equivalent to 7x + 5x + 4.
A. (7 + 5)x + 4
B. 12x + 5
C. 16x
D. 12x + 4

09 / Multiple select

Select all expressions equivalent to 4x + 4x + 4.
A. 12x
B. 8x + 4
C. (4 + 4)x + 4
D. 8x + 5

10 / Constructed response

Simplify 7(x + 2) + 7x. Explain which terms combine.

Day 057 | Unit 04 | Answer key

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1. 9x

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

2. 7x

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

3. 7x

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

4. 8x + 7

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

5. 9x + 8

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

6. 7x + 7

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

7. 5x + 10

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

8. A, D - (7 + 5)x + 4; 12x + 4

Combine coefficients of like x terms: 7 + 5 = 12. The constant 4 stays separate.

6.EE.A.3 | Multiple select

9. B, C - 8x + 4; (4 + 4)x + 4

Combine coefficients of like x terms: 4 + 4 = 8. The constant 4 stays separate.

6.EE.A.3 | Multiple select

10. 14x + 14

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response